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SARDAR PATEL UNIVERSITY
M. Sc (Information Technology)- FIRST Semester (NC)
PS01CINT54 : Relational Database Management Systems
15th April 2025, Tuesday

Time: 10.00 am to 1.00 pm

Marks:70

Q-1 Select an appropriate option for each of the following statement:

[08]

- 1 A relation will be in _____ and all non-key attributes are fully functional dependent on the primary key.
[a] 2NF if it is in 1NF [b] 3NF if it is in 2NF [c] BCNF if it is in 3NF [d] None of given
- 2 _____ represents a real-world object or concept.
[a] Entity [b] Attribute [c] Relationship [d] None of given
- 3 Multivalued attributes are depicted by _____
[a] ellipse [b] double ellipse [c] dashed ellipse [d] None of given
- 4 A _____ is a relationship in which each instance of one entity is related to many instances of another entity.
[a] many-to-one [b] one-to-many [c] one-to-one [d] None of given
- 5 _____ command is used to change values of records in a table.
[a] ALTER TABLE [b] INSERT [c] UPDATE [d] None of given
- 6 MOD(11,2) returns _____
[a] 2 [b] 1 [c] 0 [d] None of given
- 7 _____ constraint establishes uniqueness of records in a table.
[a] Primary key [b] Foreign key [c] Check [d] None of given
- 8 A _____ is a database object that holds user data.
[a] FORM [b] TABLE [c] QUERY [d] None of given

Q-2 Answer the following questions in brief. (Attempt Any SEVEN)

[14]

- 1 State at least FOUR advantages of Database Management System.
- 2 What you mean by Logical Data Independence?
- 3 Differentiate clearly Simple attribute and Composite attribute.
- 4 When you can say database is in 3NF?
- 5 Explain briefly the concept of View.
- 6 Explain with example the built in function ROUND().
- 7 Write SQL command to add primary key TNO in an existing table TELE.
- 8 Write SQL command to destroy a table EMP.
- 9 Explain briefly the concept of Trigger.

C.P.T.O.)

Q-3 Do as directed.

- [A] Explain briefly the role of different users associated with database management systems. [06]
[B] Write a note on Data Protection. [06]

OR

- [B] Explain briefly the types of Data models. [06]

Q-4 Do as directed.

- [A] Write short note on E-R Diagram. [06]
[B] Explain briefly any FOUR Codd's rules. [06]

OR

- [B] Discuss briefly the concept of Normalization. [06]

Q-5 Do as directed.

- [A] Write SQL command for each of the following. [Any FOUR] [06]
Assume that the table TELE has fields (TNO number(8,0), CNAME varchar2(15), NCALLS Number(6,0) and TOWN varchar2(15))
i. To create a table TELE having primary key TNO.
ii. To display unique records of a table TELE.
iii. To display TOWN and TOWN wise total NCALLS from table TELE.
iv. To delete all the records from a table TELE having TNO greater than 300.
v. Retrieve all information of table TELE in descending order of TNO.
[B] Explain briefly following SQL statement with example: [06]
a. DELETE statement b. ALTER TABLE Statement

OR

- [B] Explain briefly following built-in functions with example(s): [06]
a. TRUNC() b. SUBSTR()

Q-6 Do as directed.

- [A] Write short note on exception handling in PL/SQL. [06]
[B] What you mean by stored function? Explain the stored function by taking proper example. [06]

OR

- [B] List out the advantages of PL/SQL over SQL. [06]

